

# ZEWEI REN

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**Summary:** Statistics Ph.D. candidate at the University of Missouri expecting to graduate in Dec. 2026, with research experience in survival analysis, longitudinal data analysis, and multi-state modeling. Completed a summer biostatistics internship at Novartis and currently works as a research assistant providing statistical consulting support through the Center for Applied Statistics and Data Analysis (CASDA). Proficient in R, SAS, and Python, seeking biostatistician, statistician or related roles.

## EDUCATION

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**University of Missouri, Columbia, MO** Aug. 2021 – Expected Dec. 2026  
Ph.D. in Statistics

**University of Connecticut, Storrs, CT** Aug. 2019 – Jun. 2021  
M.S. in Statistics

**Central South University, Changsha, China** Sep. 2014 – Jun. 2018  
B.S. in Statistics

## TECHNICAL SKILLS

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**Programming:** R, SAS, Python, JAGS

**Statistical Methods:** Survival Analysis, Longitudinal Data Analysis, Multi-State Models, Frailty Models, Mixture Models, High-Dimensional Data Analysis, Transfer Learning, Bayesian Method

**Tools:** L<sup>A</sup>T<sub>E</sub>X, Microsoft Office (Excel, Word, PowerPoint), HPC Parallel Computing, Codex CLI/IDE(Cursor)

## WORK EXPERIENCE

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**Research Assistant** Aug. 2022 – Present  
Center for Applied Statistics and Data Analysis (CASDA), University of Missouri *Columbia, MO*

- Consult with faculty, students, and external collaborators to define research goals and develop appropriate statistical analysis plans.
- Apply advanced statistical methods, including generalized linear models, mixed-effects models, survival models, power analysis, and study design, in interdisciplinary projects.
- Provide R programming support for research and dissertation projects, and deliver office hours and workshops, including the R workshop for ASA DataFest at Mizzou.

**Biostatistician – Summer Intern** Jun. 2025 – Aug. 2025  
Novartis *East Hanover, NJ*

- Collaborated with trial statisticians and cross-functional partners to develop and implement an interval-censored multi-state modeling framework for longitudinal clinical trial data with repeated LDL-C measurements, improving estimation accuracy and informing observation-interval decisions in trial design.

**Statistician** Jan. 2022 – Aug. 2022  
School of Medicine, University of Missouri *Columbia, MO*

- Supported MU Hospital research projects by collaborating with physicians, conducting statistical analyses, improving R workflows, and reviewing domain-specific literature to refine research questions.

**Data Analyst Intern** Jan. 2019 – Feb. 2019  
Strategical Investment Department, Cameron Pace Group, Beijing Branch *Beijing, China*

- Collected, visualized, and analyzed data on mergers, acquisitions, and equity investments in the Technology, Media, and Telecommunications(TMT) and glass-free 3D film industries to support investment decision-making.

## RESEARCH EXPERIENCE

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### Subgroup Analysis for Interval-Censored Data with a Cure Fraction by a Mixture Promotion Time Cure Model

Feb. 2024 – Present

Advisor: Dr. Jianguo Sun

University of Missouri

- Developed a mixture promotion time cure model for interval-censored data with a cure fraction to capture heterogeneous covariate effects on both cure probabilities and survival outcomes.
- Proposed a subgroup analysis framework using the BIC criterion to determine the number of subgroups, and developed a computationally efficient variance estimation procedure based on profiled likelihood and the delta method.

### Variable Selection and Regression Analysis for Clustered Mixed Panel Count Data with $L_0$ Seamless Penalty

Jan. 2025 – Present

Advisor: Dr. Jianguo Sun

University of Missouri

- Developed a unified modeling framework for clustered mixed panel count data that accommodates recurrent event counts and simultaneously observed binary outcomes, while addressing within-cluster correlation to mitigate inefficiency and bias.
- Integrated variable selection and regression modeling using the minimum information criterion (MIC), a seamless  $L_0$  penalty, to identify relevant covariates without tuning parameters, achieving consistent and computationally efficient estimation.

### Application of Multi-state model with Interval Censoring on Longitudinal Data for Optimal Observation Interval Selection in Trial Design Phase

Jun. 2025 – Sep. 2025

Mentor: Dr. Jahangir Alam

Novartis

- Implemented a time-inhomogeneous interval-censored multi-state model with piecewise-constant baseline hazard approximation, demonstrating through simulation and Inclisiran trial data that it improves estimation accuracy and guides optimal observation interval in clinical trials design.

### Bayesian Hierarchical Modeling for Longitudinal PBC Data

Oct. 2022 – Dec. 2022

Course Project, Bayesian Analysis

University of Missouri

- Applied Bayesian hierarchical longitudinal modeling using R/JAGS to analyze Mayo Clinic PBC data, comparing fixed-effect, random-intercept, and random-slope models with frequentist alternatives and estimating patient-specific bilirubin trajectories via MCMC.

### Experiment Design & Efficacy Analysis (Phase II)

Aug. 2020

Mentor: Dr. Shoubhik Mondal

UConn Biopharmaceutical Summer Academy with Boehringer Ingelheim

- Calculated necessary sample size using Hypothesis Testing and MCP-Mod method given assumed effect size.
- Used Fisher exact test to analyze significant differences of dose effect, and applied ANCOVA model and Logistic Model in detecting significant effects of baseline characteristics on primary and secondary endpoints respectively.

## PUBLICATIONS

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Sun, Jianguo, Yuxiang Wu, Mingyue Du, and **Zewei Ren**. "A Mixture Model Approach for Subgroup Analysis of Interval-Censored Failure Time Data." *Available at SSRN 5244196*.

Pérez-López, Dennis O., Matthew J. Burke, Chady H. Hakim, James A. Teixeira, Jin Han, Yongping Yue, **Zewei Ren** et al. "Circulatory CCL2 distinguishes Duchenne muscular dystrophy dogs." *Disease Models & Mechanisms* 18, no. 3 (2025).

Anbari, Allison B., Knoo Lee, Jennifer M. Hulett, Lada Micheas, **Zewei Ren**, Anthony Belenchia, and Suhwon Lee. "Exploring disparities in post-cancer treatment instructions: an analysis of rural vs. urban breast cancer survivors in Missouri using BRFSS data." *BMC Health Services Research* 24, no. 1 (2024): 1-10.